Lab 08 Manual

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SRT411NAA: Digital Data Analytics

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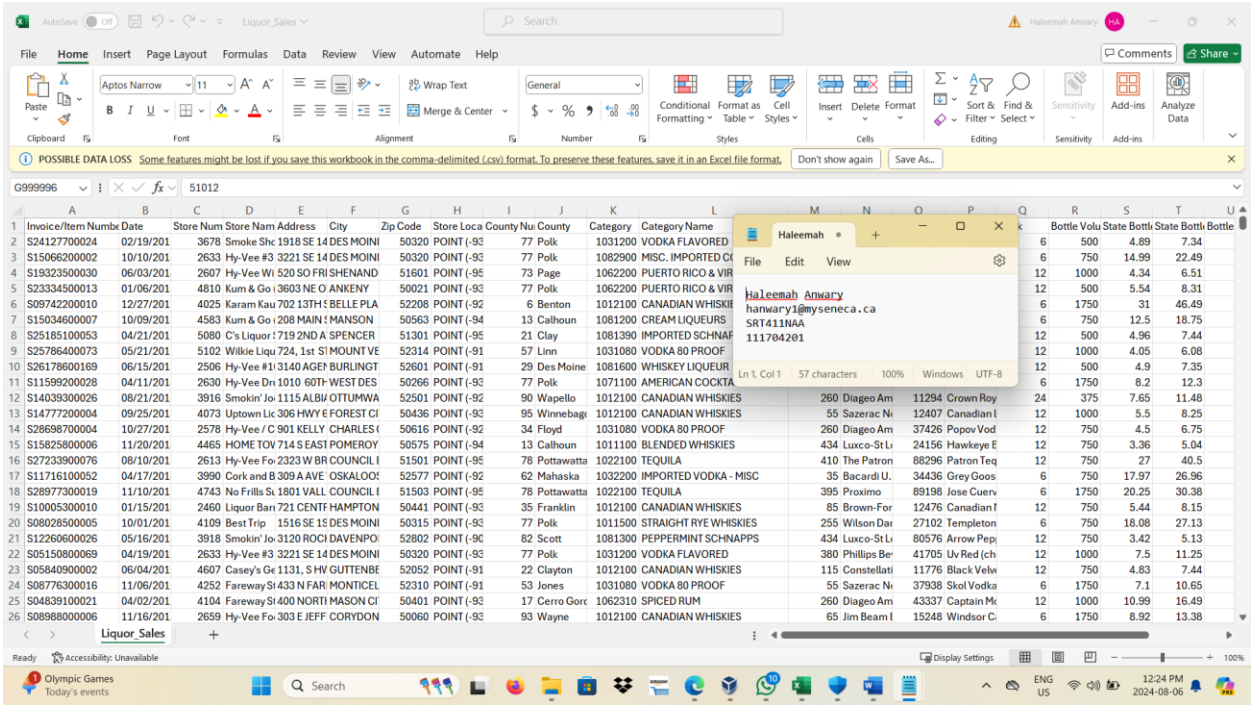
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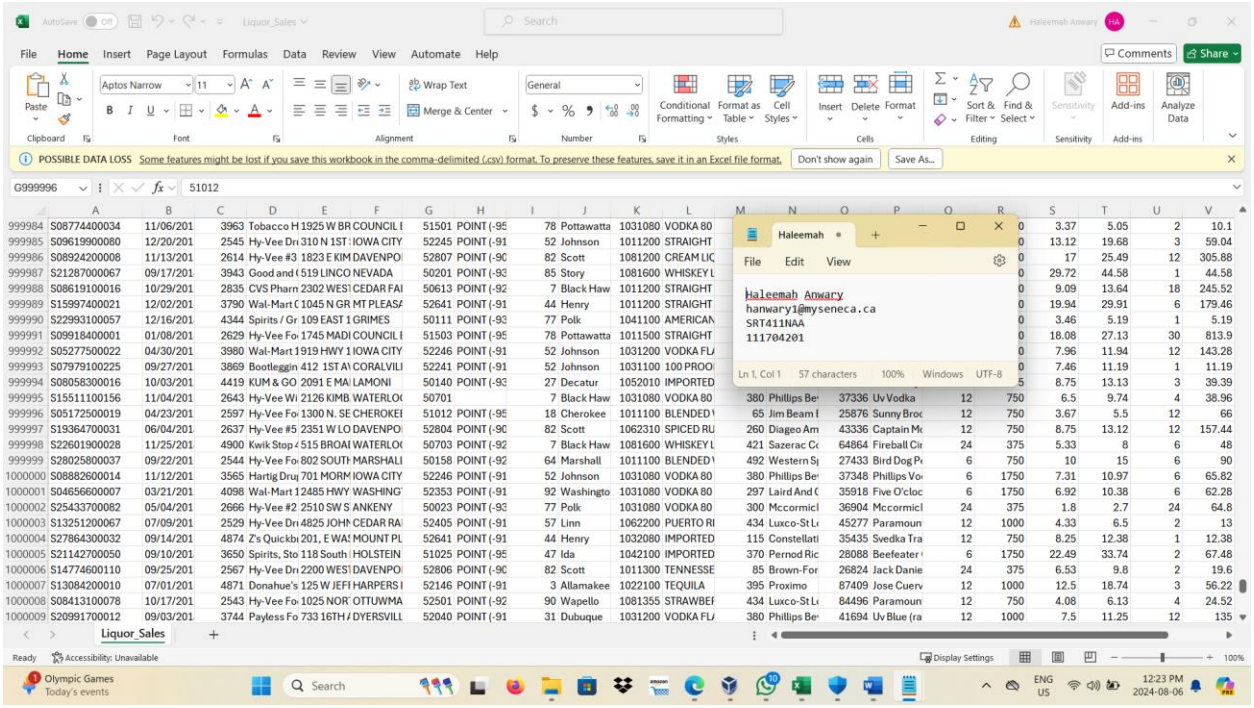
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Task 01 (Data Story and Research Question)

I chose the data Liquor_sales from Kaggle which has more than 100k rows and more than 10 columns





1) Describe what data, and data story the data demonstrates

Data Overview:
The dataset captures sales transactions for various alcoholic beverages across different stores. It includes detailed information about the sales, such as the store's location, the product sold, the volume, and the monetary value of the sales.

Potential Data Story:
The data can be used to analyze trends in alcohol sales across different regions, store types, and time periods. A potential data story could explore how sales vary by location (e.g., urban vs. rural areas), by specific products, or by time (e.g., seasonal trends). Another angle could be investigating the relationship between pricing strategies (state bottle cost and retail) and sales volume

2) Describe Variables and Dimensions

Variables:
Quantitative:

- **Store Number:** Numerical identifier for each store.
- **Zip Code:** Postal code of the store.
- **County Number:** Numerical identifier for each county.
- **Item Number:** Numerical identifier for each item.
- **Pack:** Number of items in a pack.
- **Bottle Volume (ml):** Volume of the bottle in milliliters.
- **State Bottle Cost:** Cost of the bottle at the state level.
- **State Bottle Retail:** Retail price of the bottle.
- **Bottles Sold:** Number of bottles sold.
- **Sale (Dollars):** Total sales in dollars.
- **Volume Sold (Liters):** Volume of alcohol sold in liters.
- **Volume Sold (Gallons):** Volume of alcohol sold in gallons.

Qualitative:

- **Invoice/Item Number:** Unique identifier for the transaction.
- **Date:** Date of the transaction.
- **Store Name:** Name of the store.
- **Address:** Address of the store.
- **City:** City where the store is located.
- **County:** Name of the county.
- **Item Description:** Description of the item sold.
- **Store Location:** Geographical location of the store (in point format).

Dimensions:

Geographical Dimensions:

- City, County, Zip Code, Store Location

Product Dimensions:

- Item Number, Item Description, Pack, Bottle Volume (ml)

Sales Dimensions:

- Date, Store Name, Store Number, Bottles Sold, Sale (Dollars)

3) List Data Types and Cohesively organise and clean data

Data Types:

String (Object):

- Invoice/Item Number, Date, Store Name, Address, City, County, Item Description, Store Location

Integer:

- Store Number, Zip Code, County Number, Item Number, Pack, Bottles Sold

Float:

- Bottle Volume (ml), State Bottle Cost, State Bottle Retail, Sale (Dollars), Volume Sold (Liters), Volume Sold (Gallons)

4) Design a Data Story Research Questions

Geographical Analysis:

- How do alcohol sales vary across different counties and cities?
- Are there regional trends in the types of alcohol sold?

Product Performance:

- What are the best-selling alcohol products, and how do they differ across locations?
- How do sales volumes change with different packaging (e.g., bottle volume, pack size)?

Temporal Analysis:

- Are there any seasonal trends in alcohol sales?
- How have sales trends changed over time?

Pricing and Sales:

- Is there a relationship between the state bottle cost/retail price and the volume of sales?
- How do different pricing strategies impact sales across different stores?

5) What fields are available and are relevant to the research question

Geographical Analysis:

- City, County, Zip Code, Store Location

Product Performance:

- Item Number, Item Description, Pack, Bottle Volume (ml), Bottles Sold, Sale (Dollars)

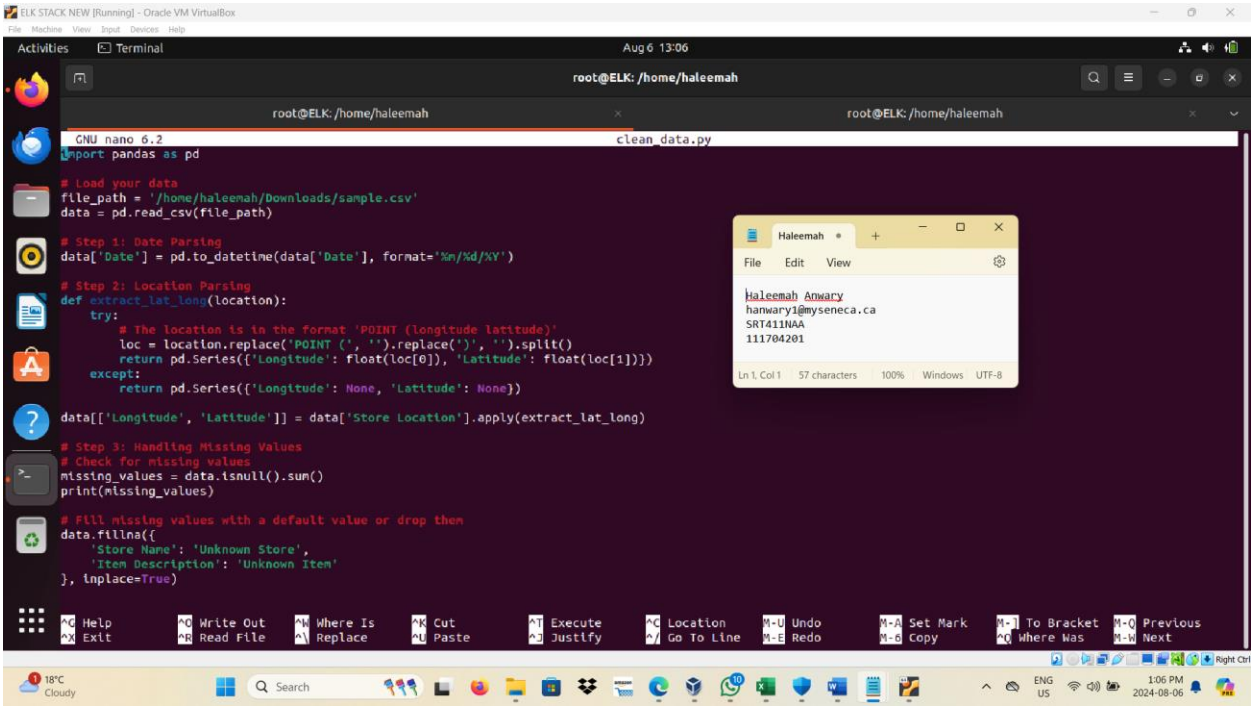
Temporal Analysis:

- Date

Pricing and Sales:

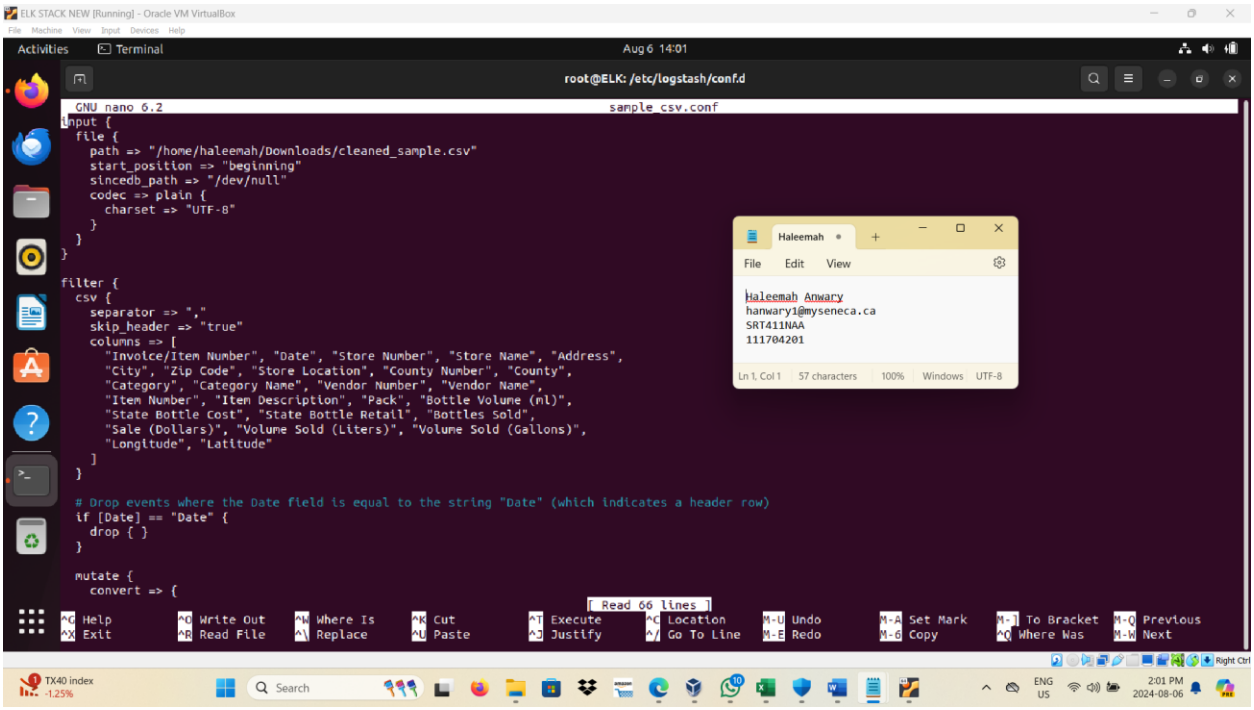
- State Bottle Cost, State Bottle Retail, Bottles Sold, Sale (Dollars)

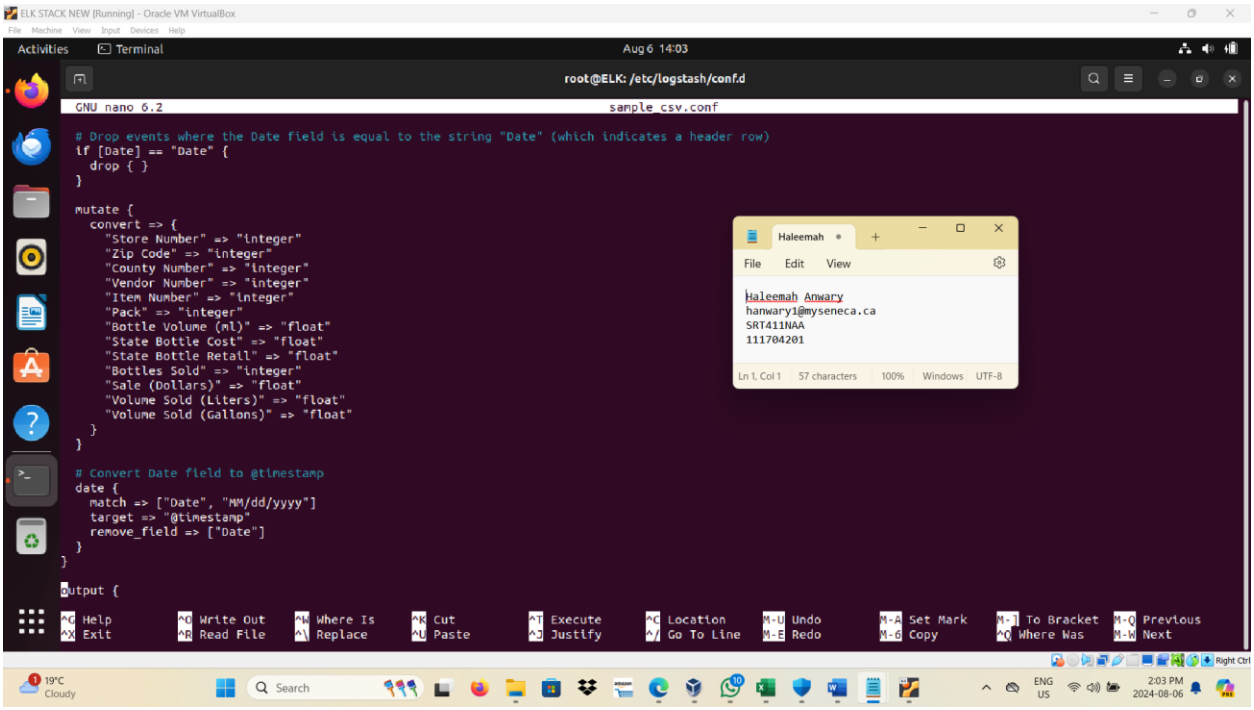
Data Cleaning: I used this Python Script to clean the data

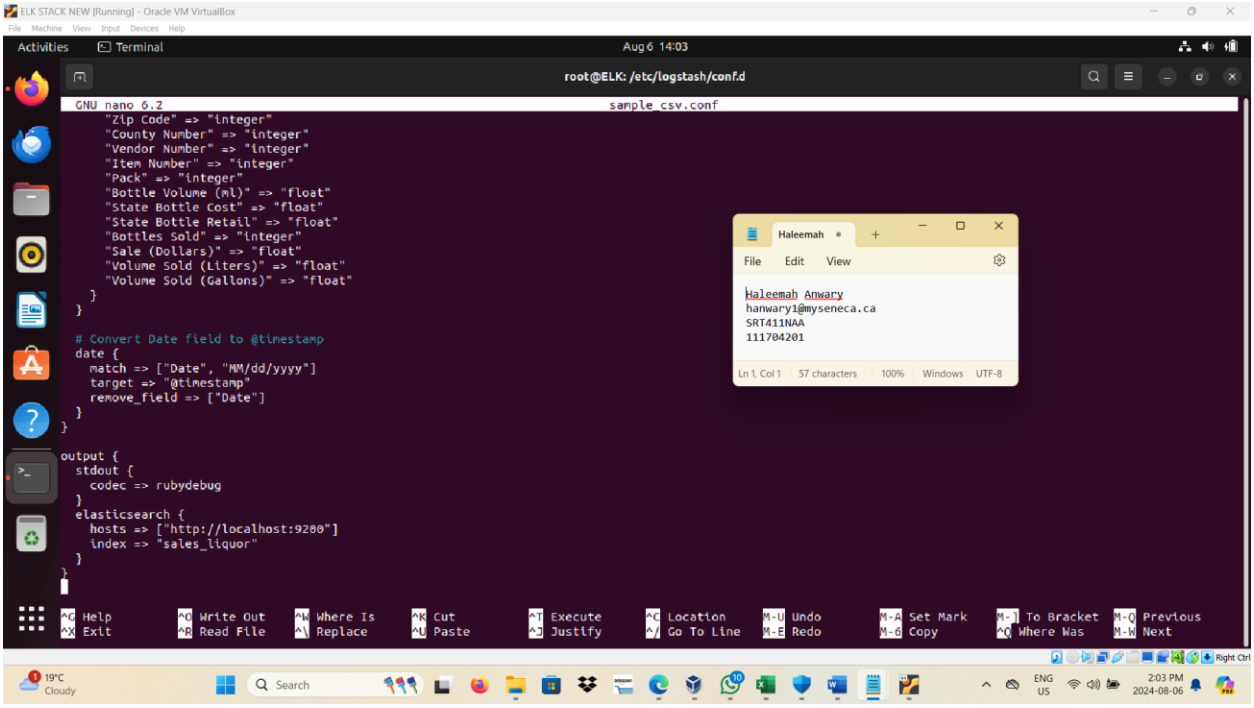


Task 02 (Data Conversion)

Logstash Config File

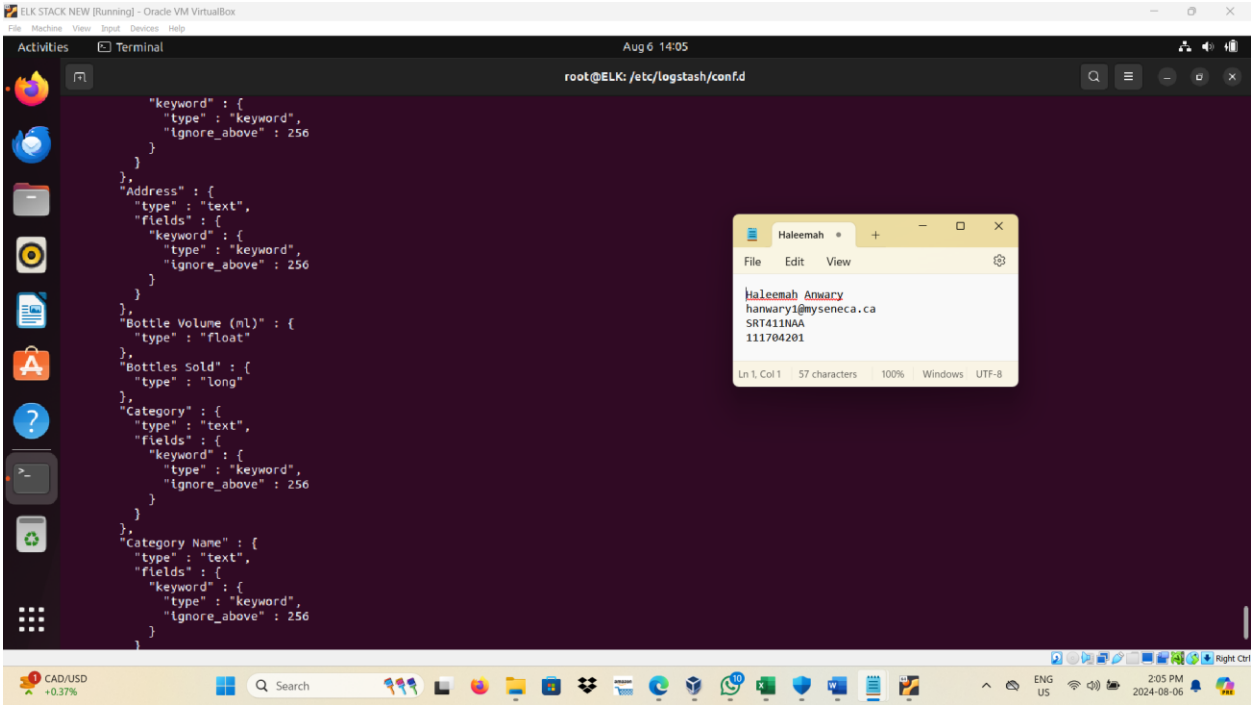




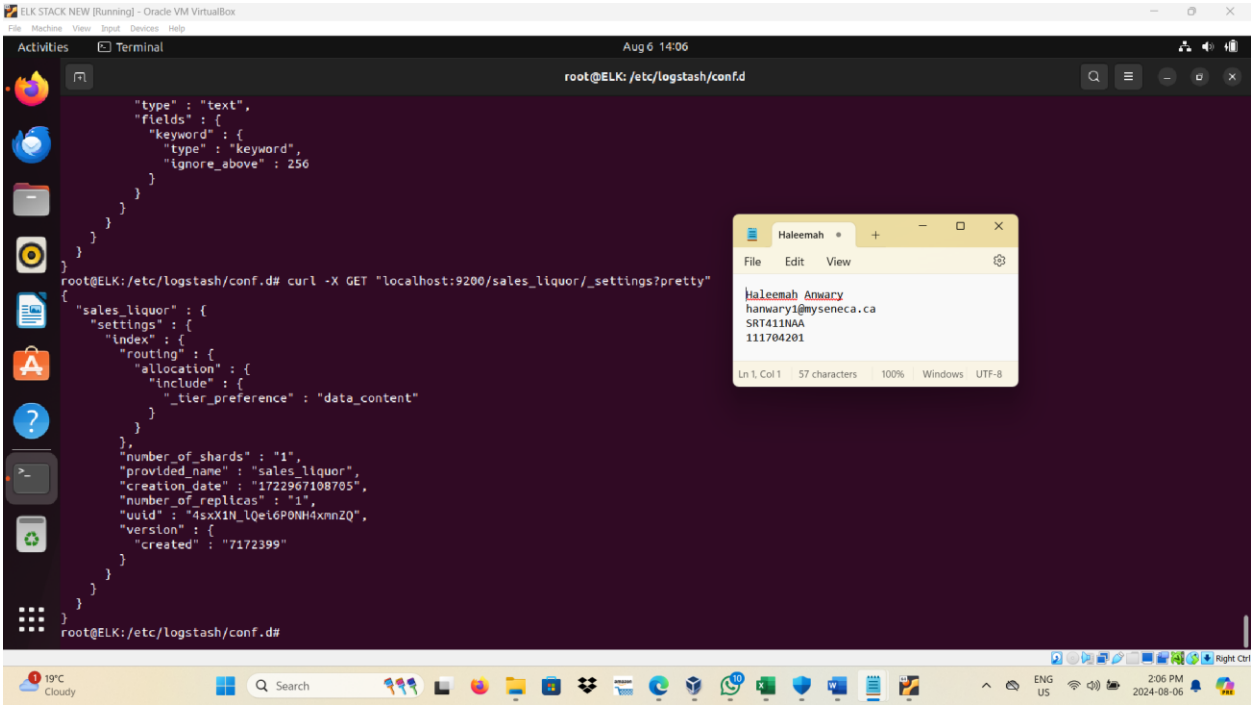


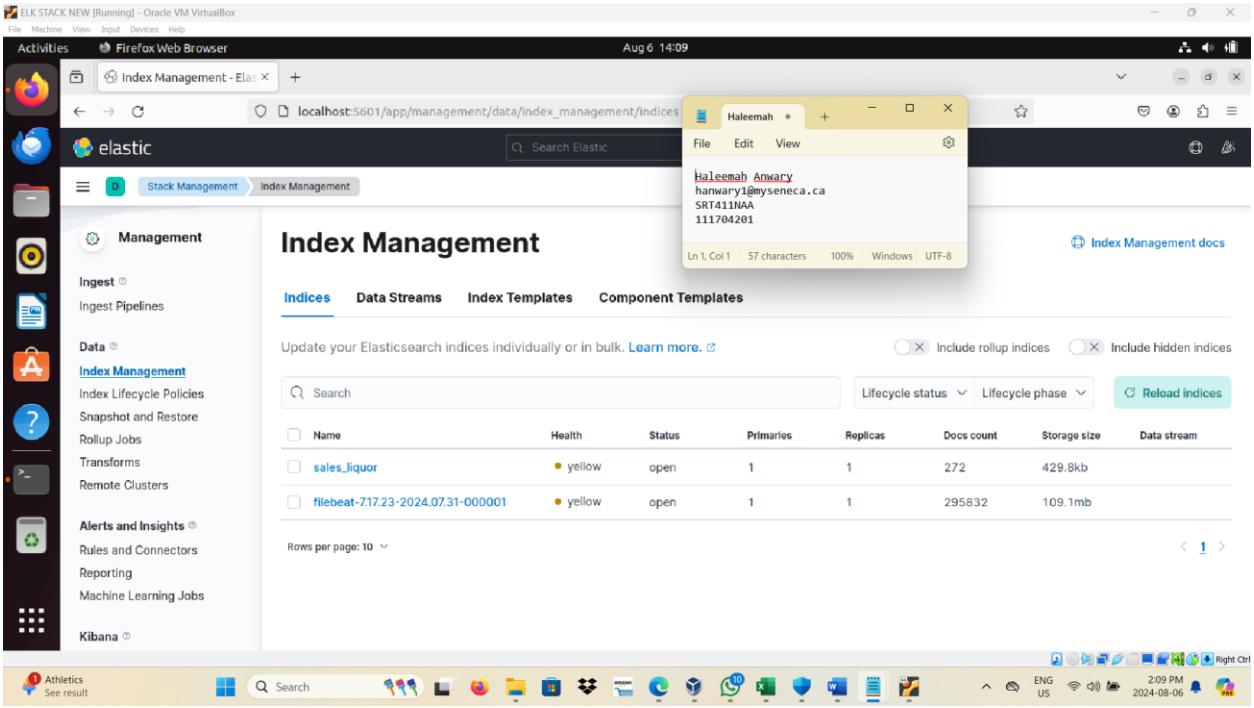
Task 03 (Data Views and Index Mapping)

Index Structure

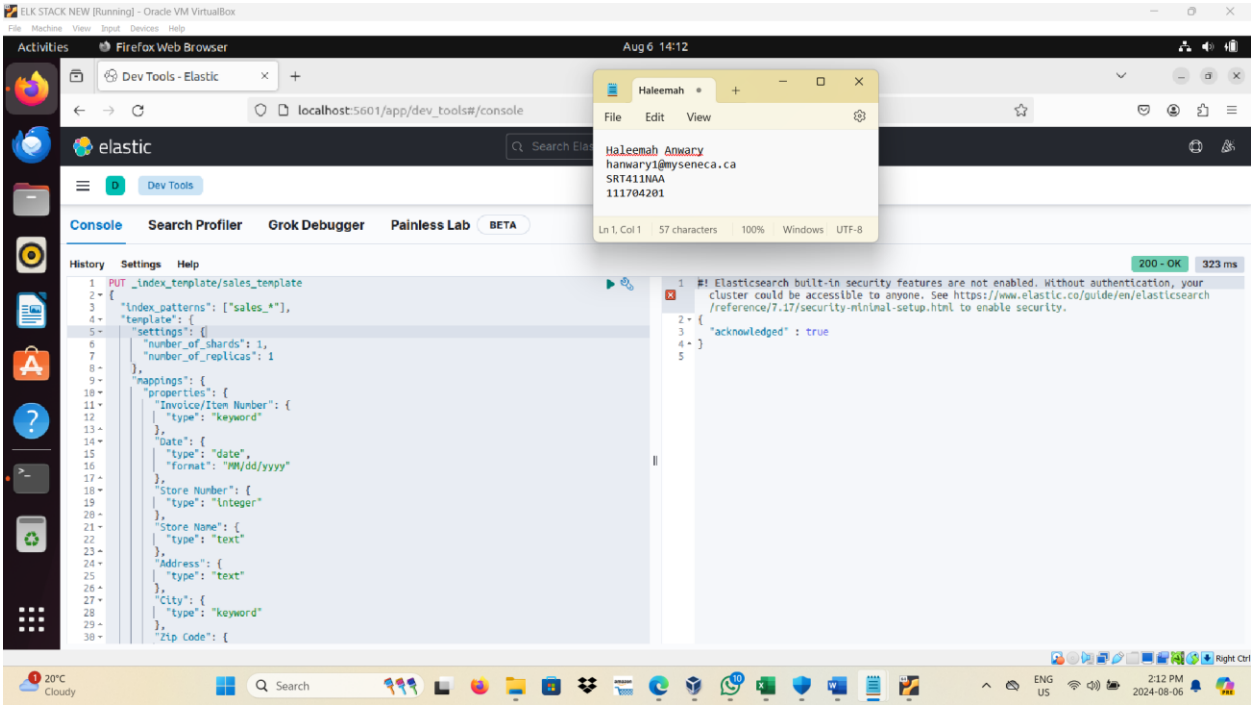


Index Settings

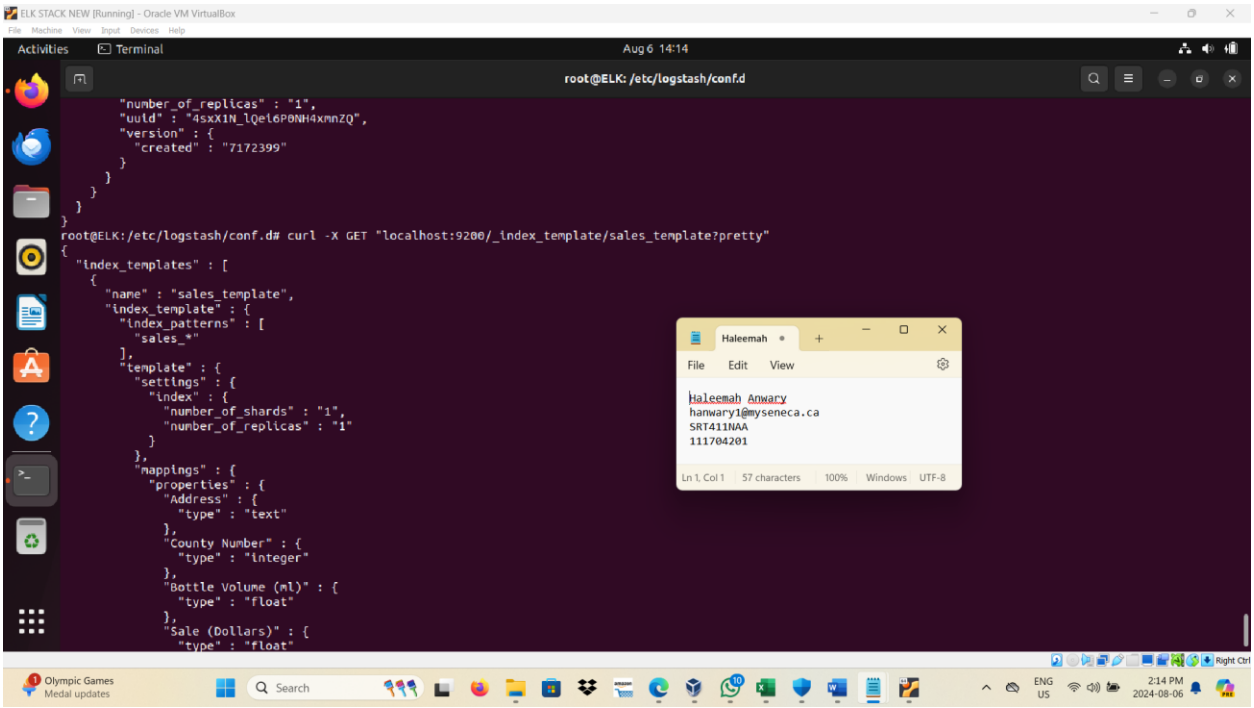




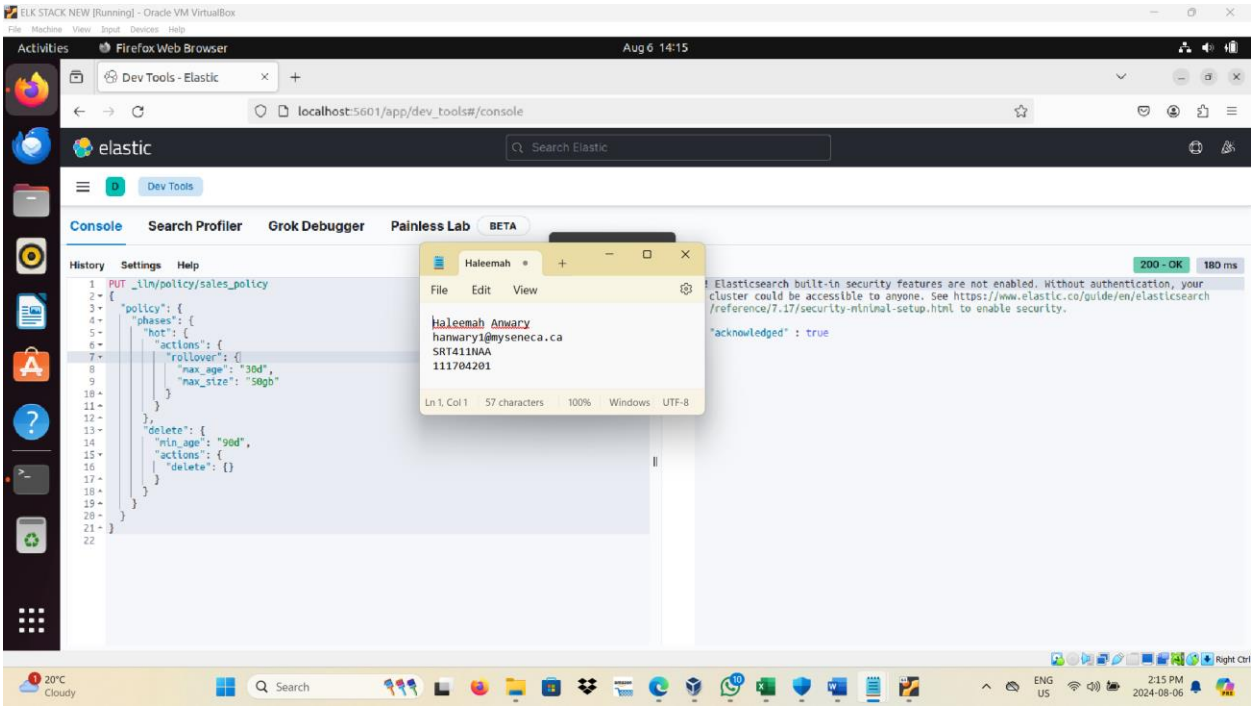
Create Index template



Verify Template

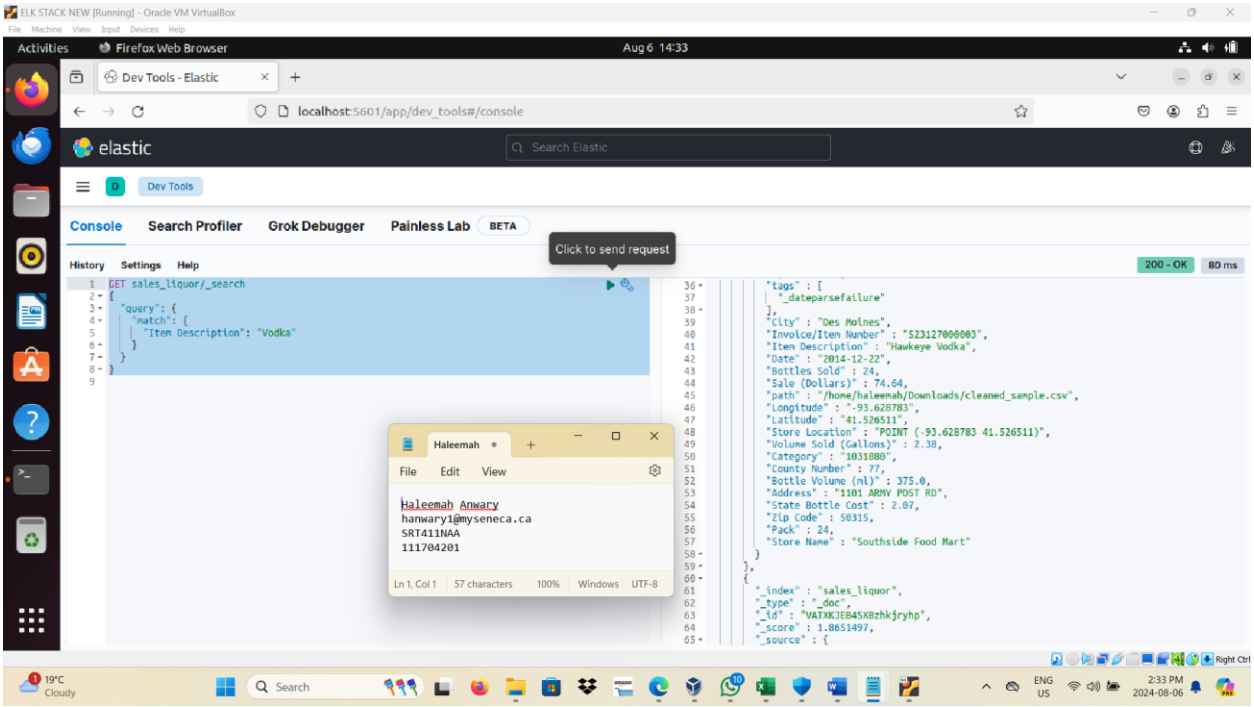


Memory Management

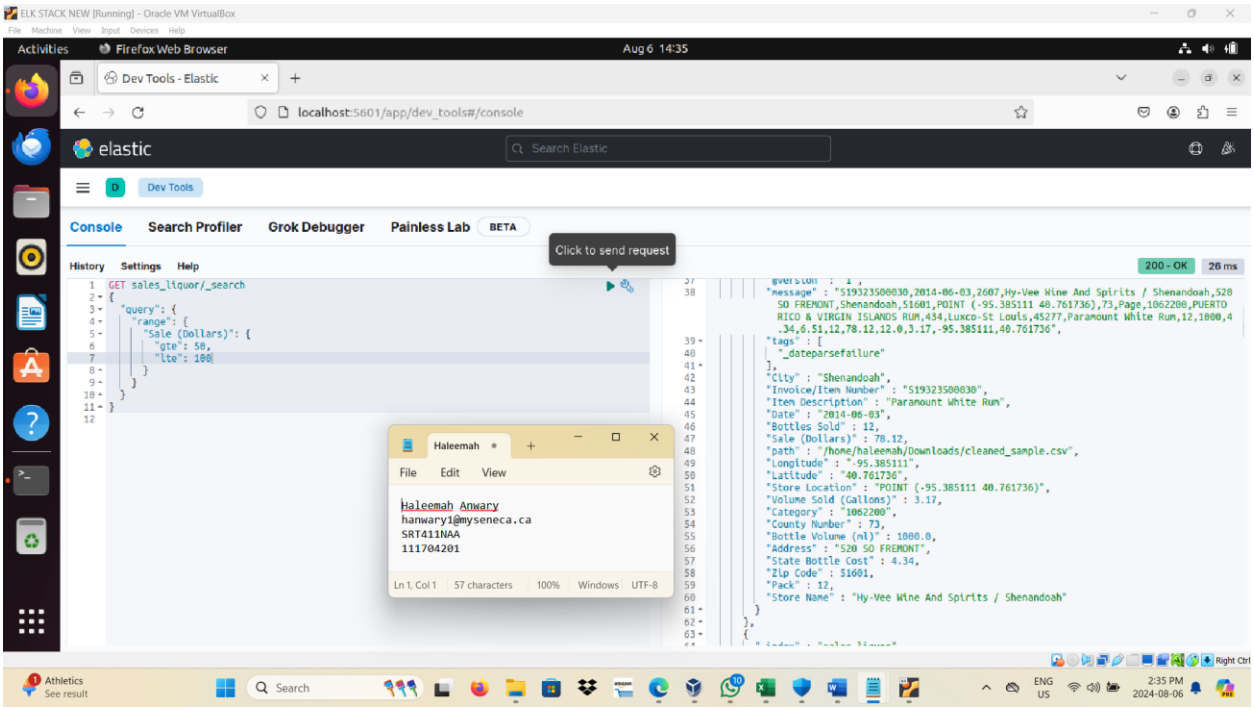


Task 04 (ELK Query)

Searches for documents where the Item Description field contains the word "Vodka".

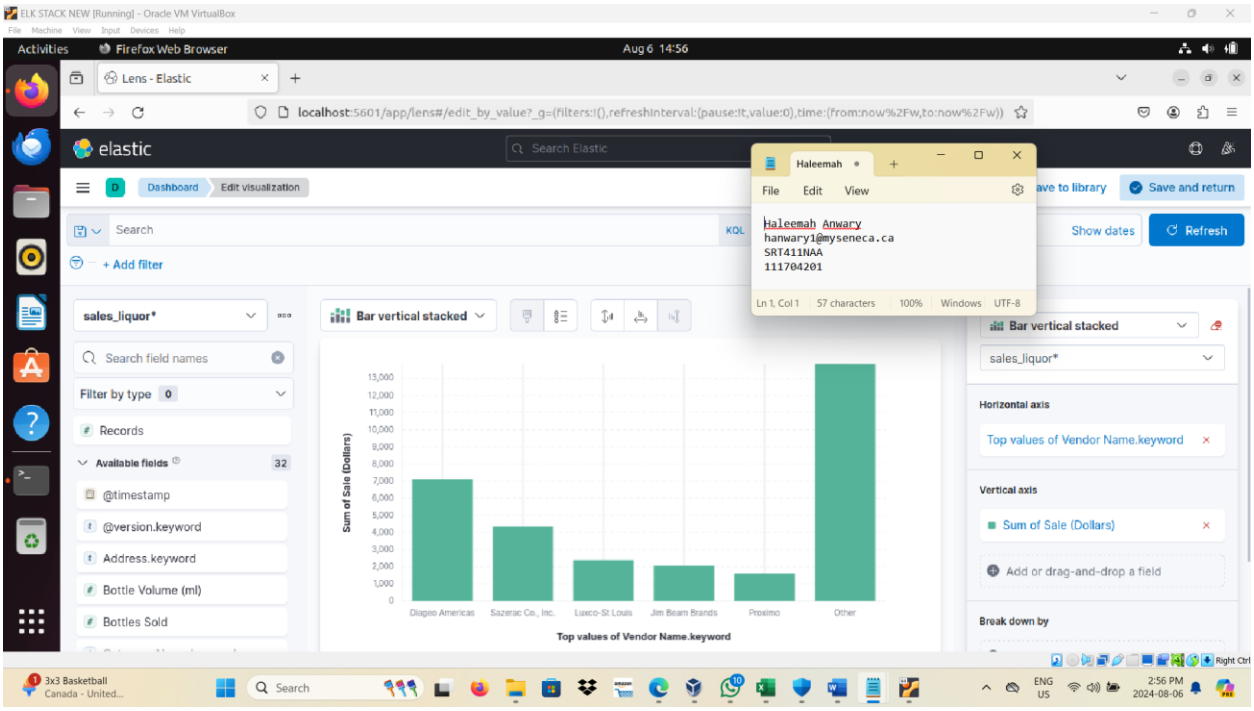


Searches for documents where the Sale (Dollars) field is between $50 and $100.

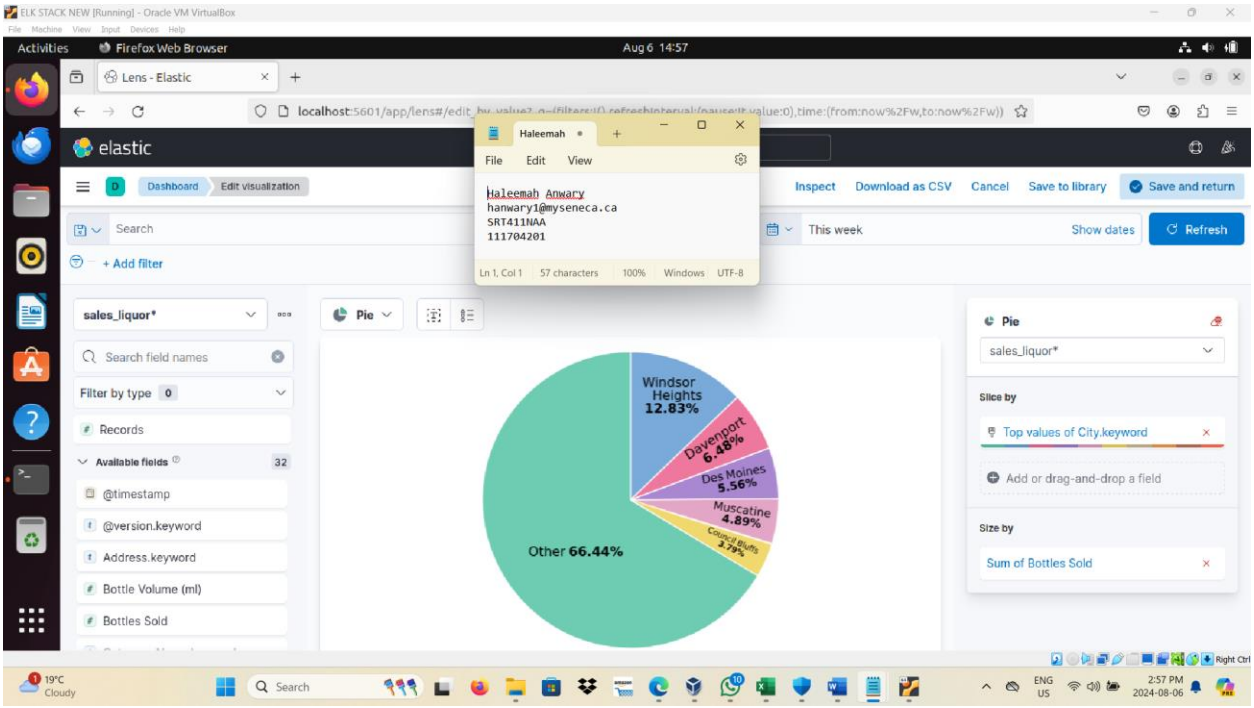


Task 05 & 06 (Data Visualization and Dashboard)

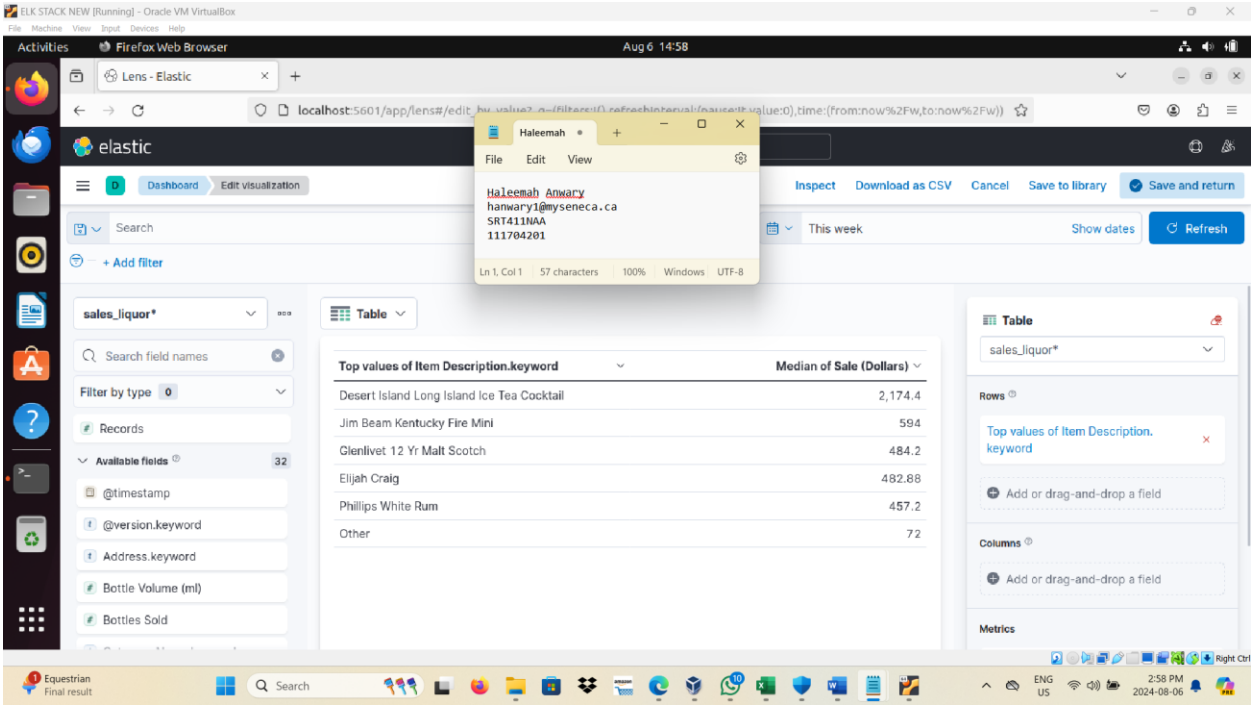
Sum of Sales by Vendor



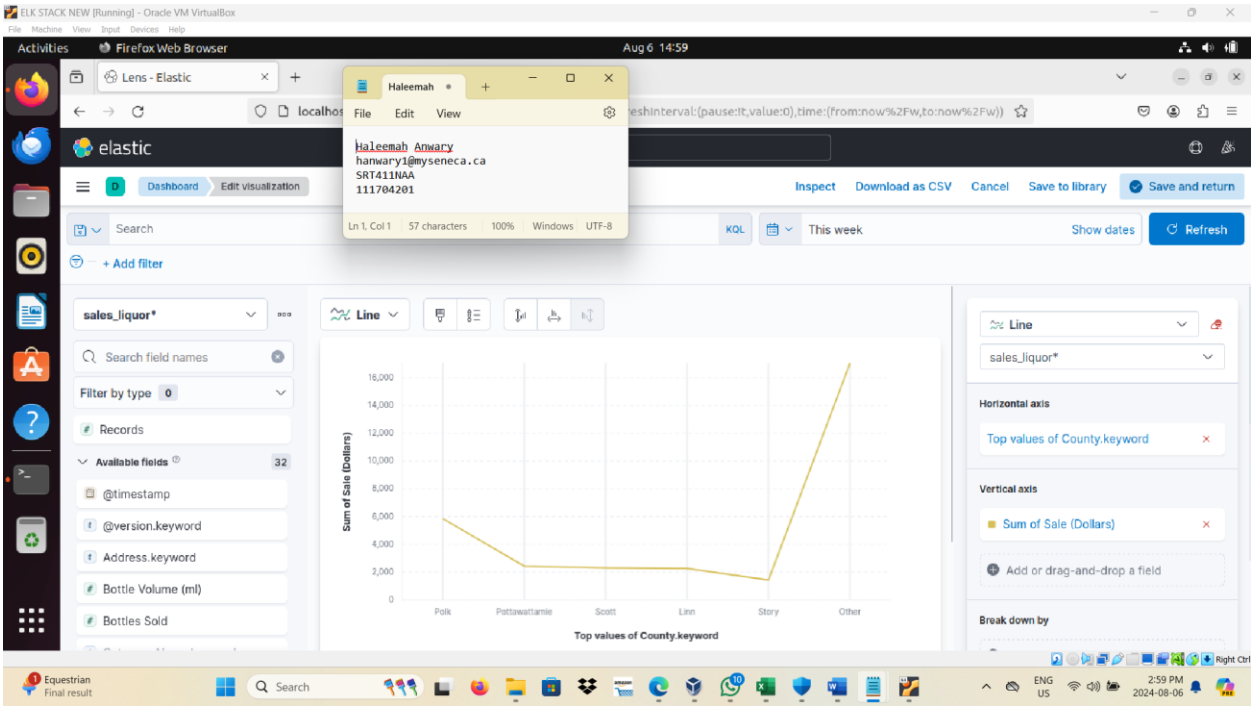
Bottles sold by City



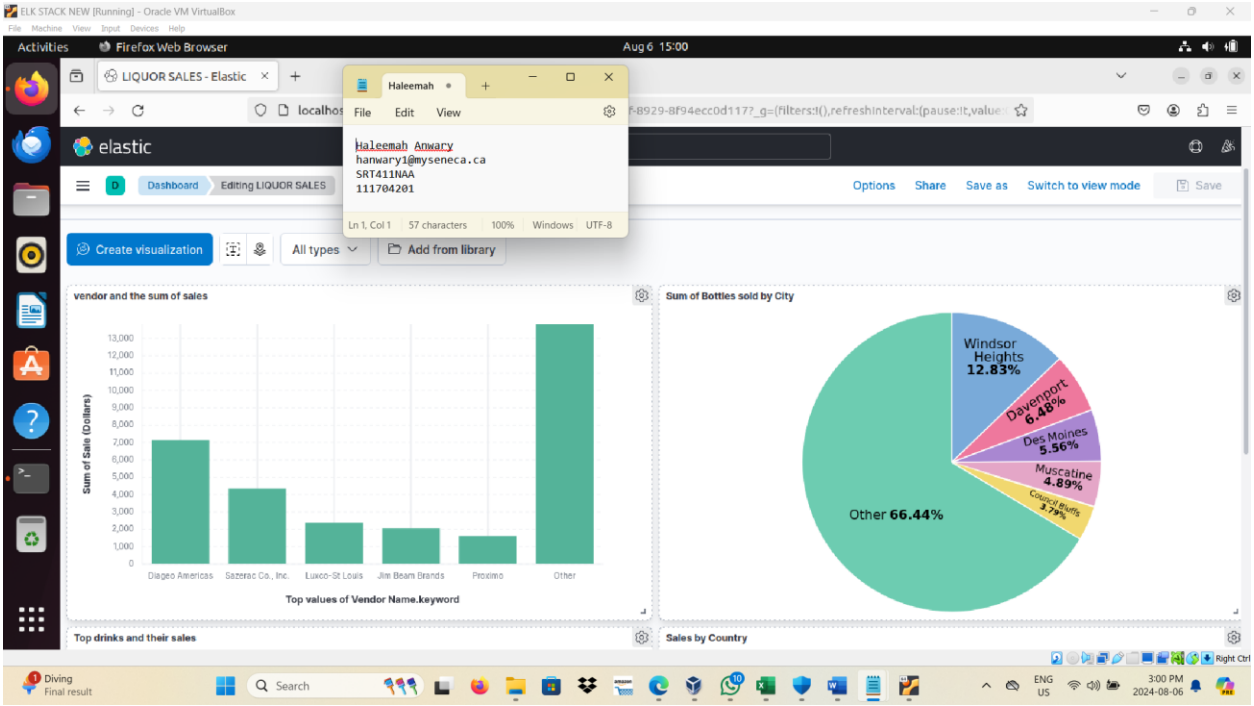
Items and their median sales

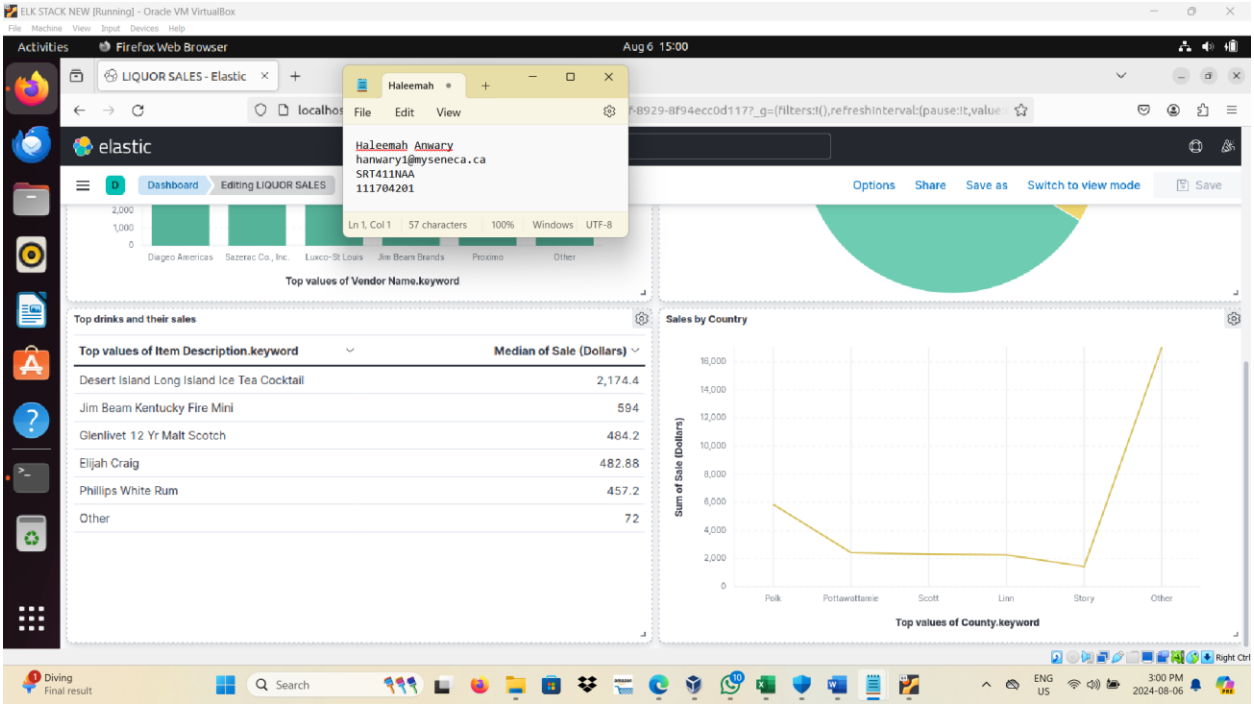


Sales by Country

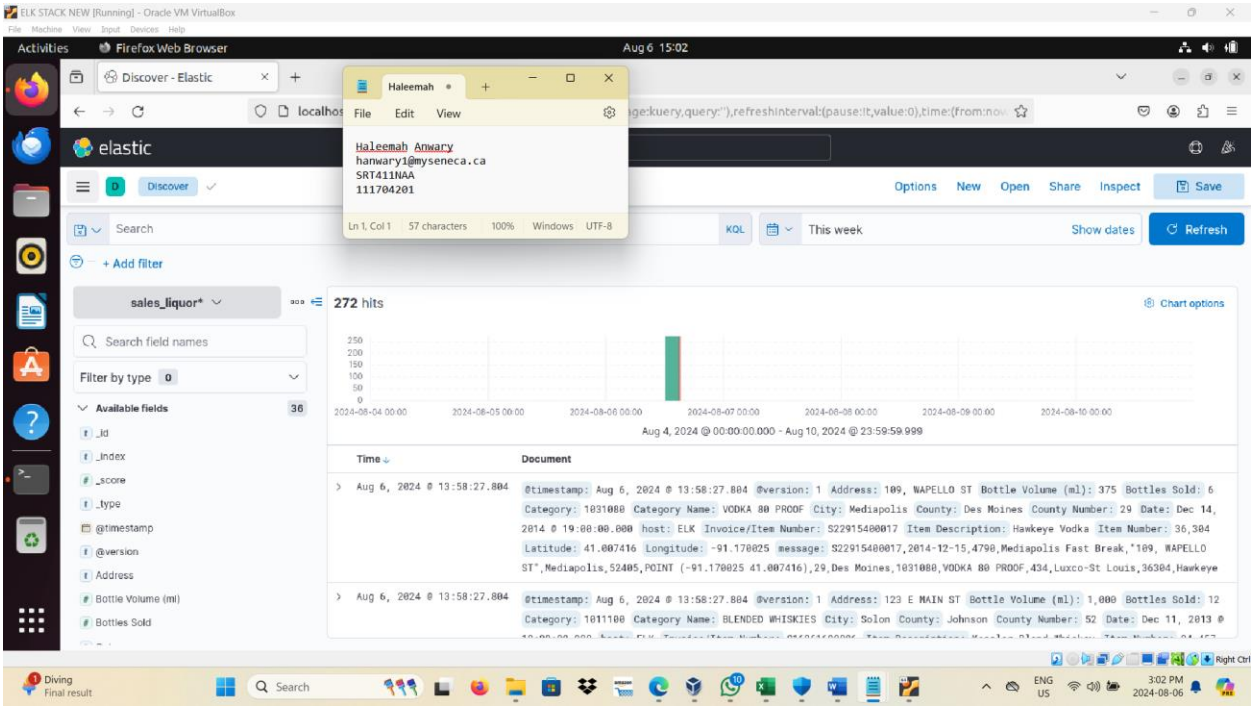


My DASHBOARD called “LIQUOR SALES”

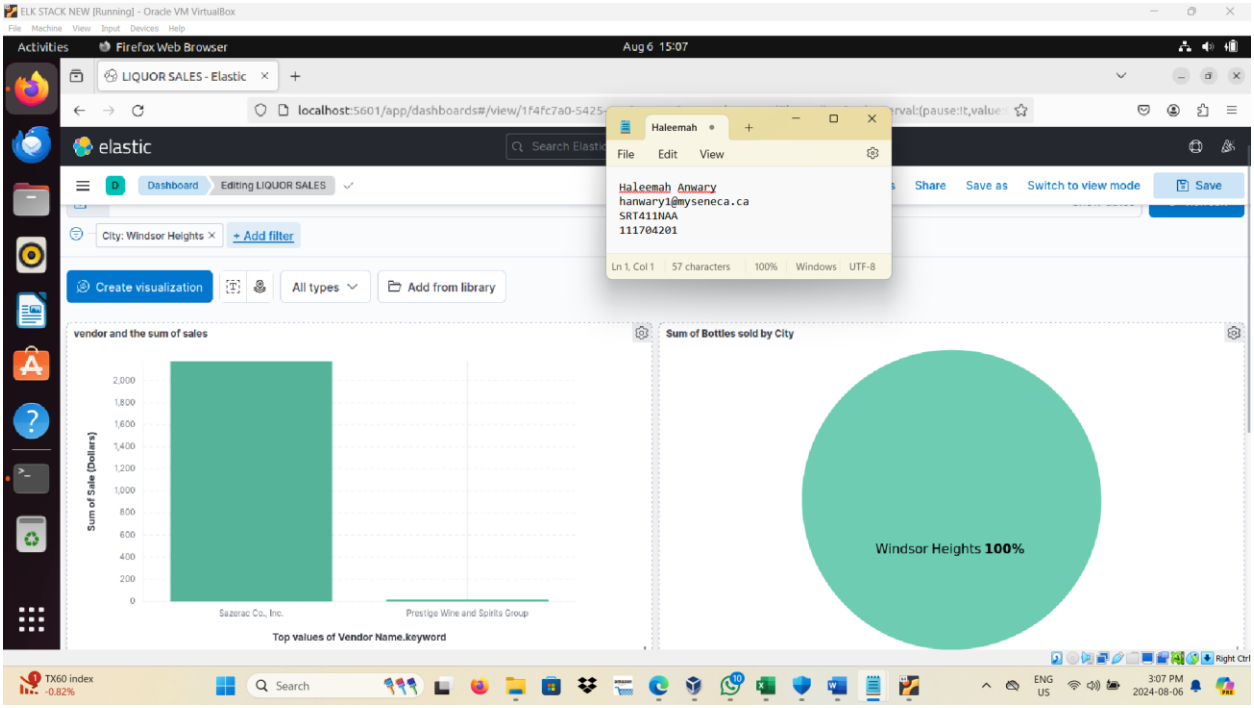


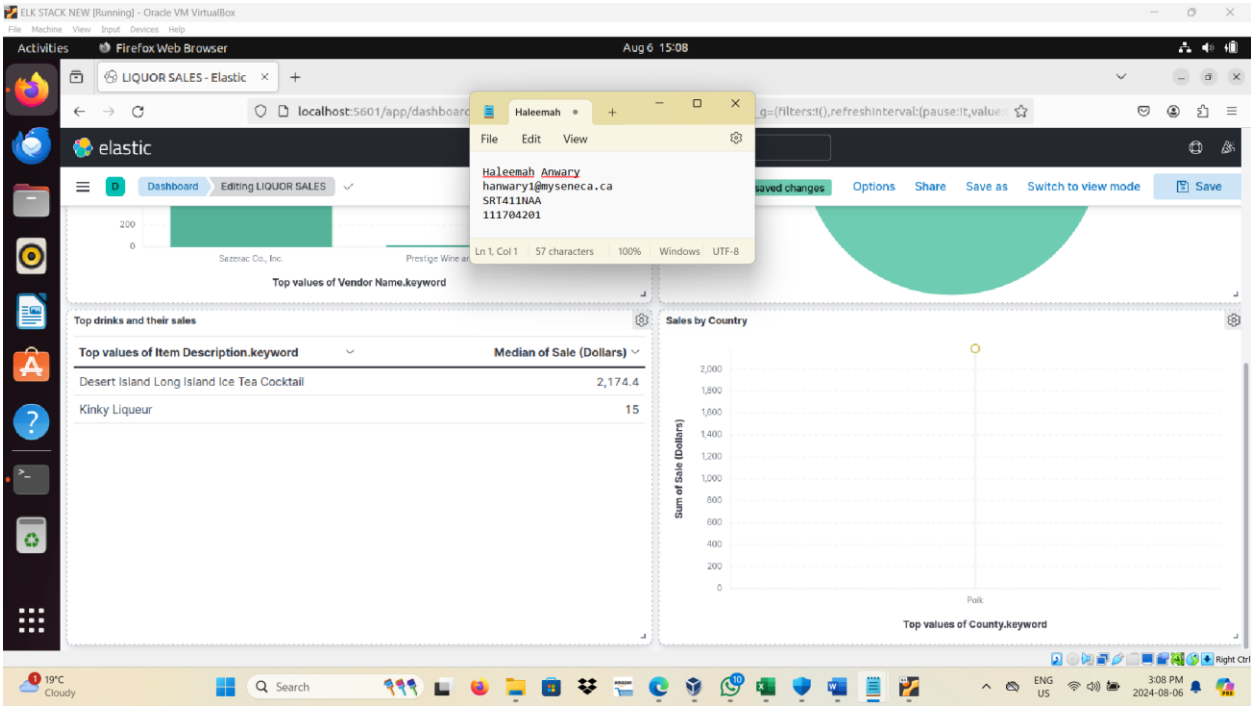


Data in DISCOVER

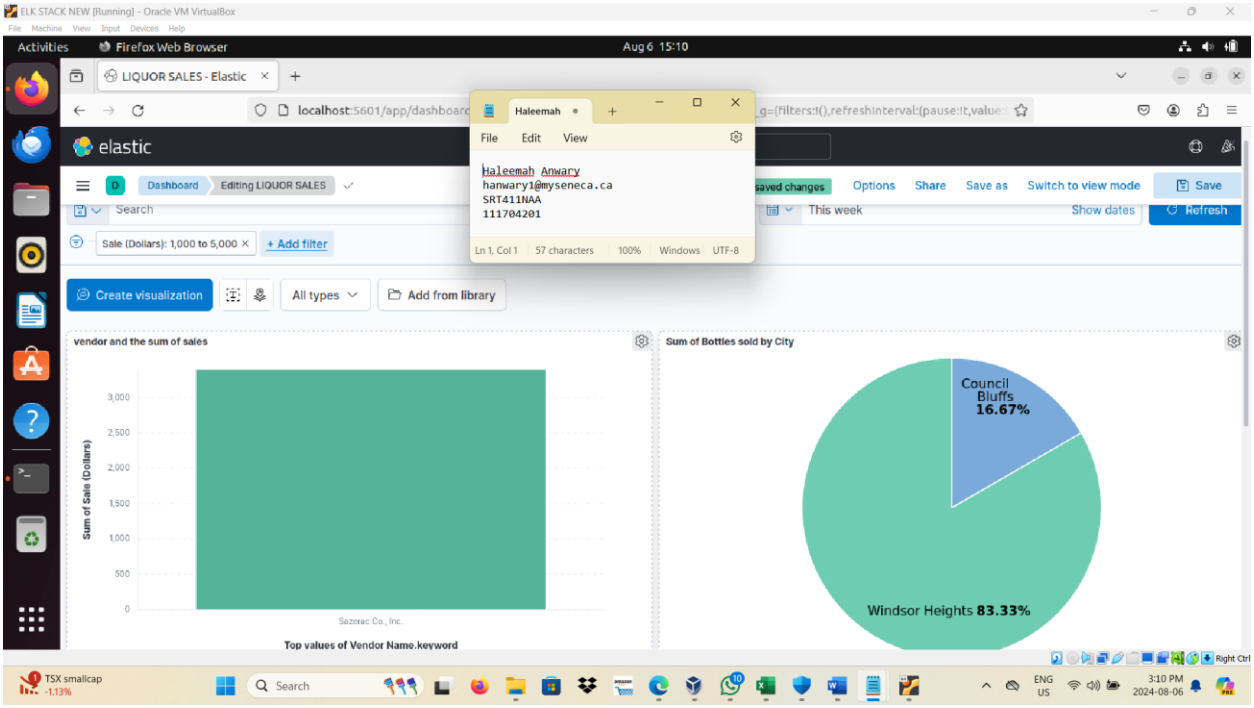


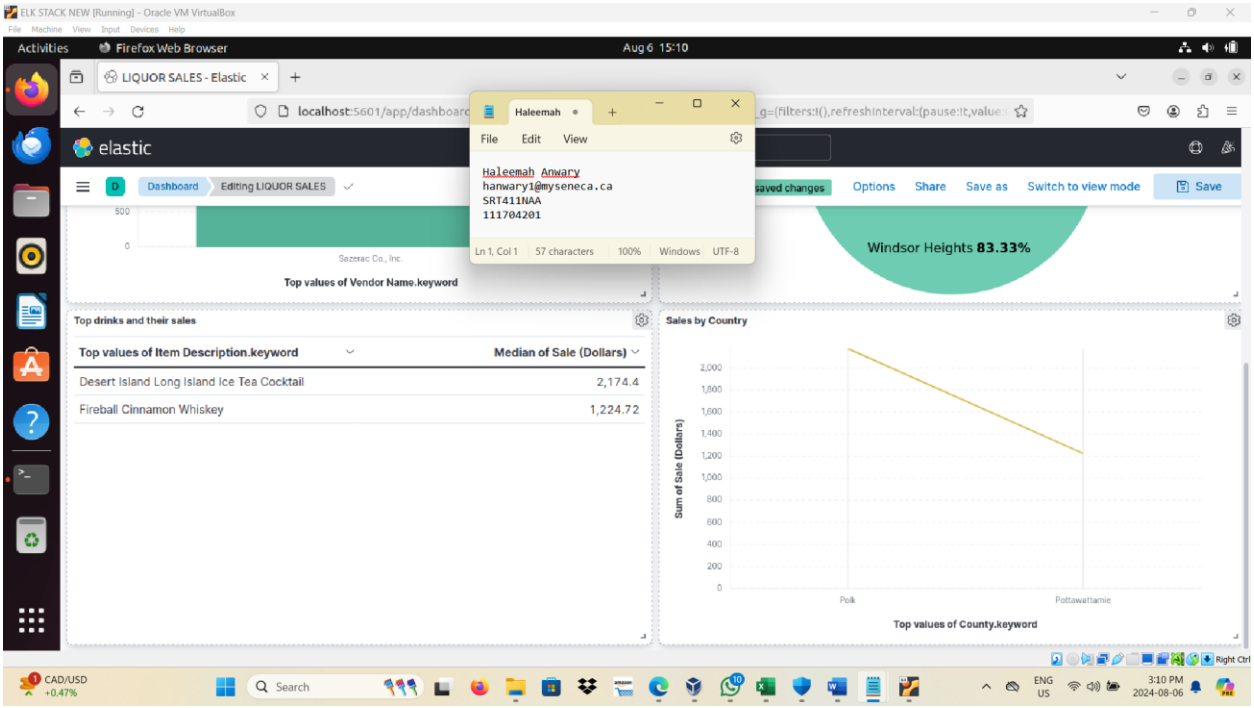
Filtering by City = Windsor Heights





Filtering by sales is between 1000-5000

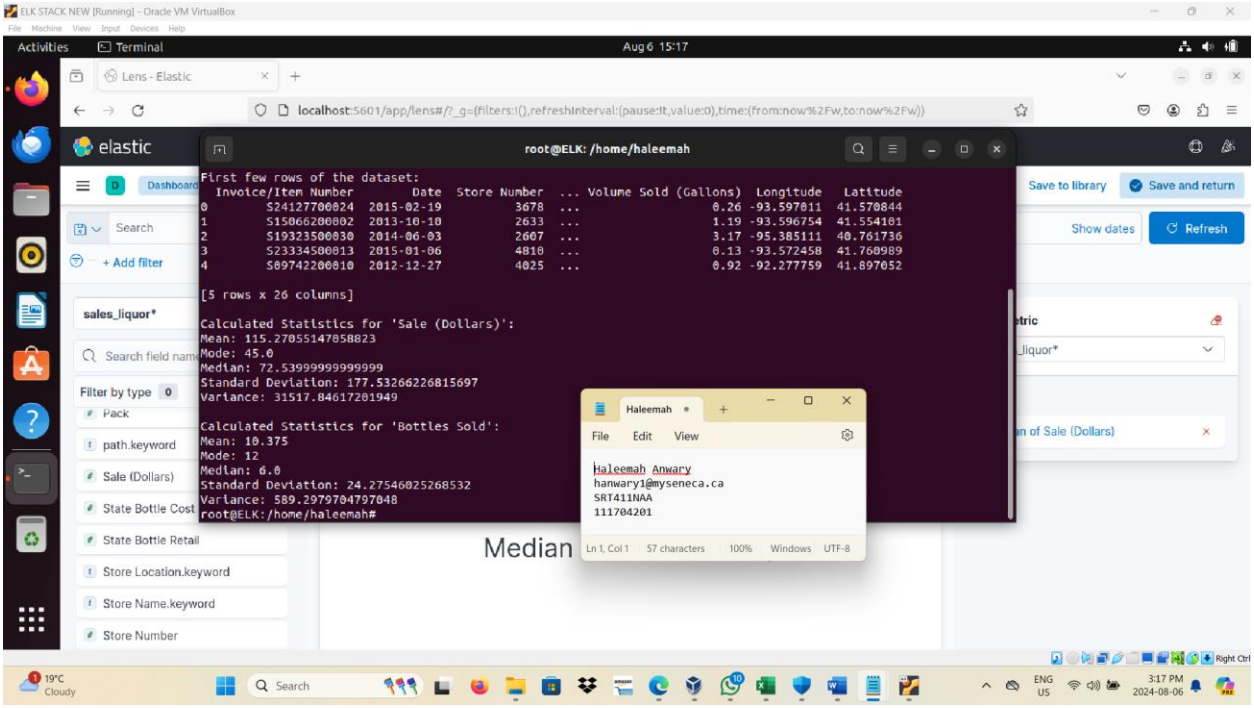




Calculate mean, mode, median, standard deviation, and variance for one or two columns from your dataset and explain their significance

Interpretation of Results:

- Mean: Represents the average sale amount.
- Mode: The most frequently occurring sale amount.
- Median: The middle value in the distribution of sales.
- Standard Deviation: Indicates the spread of sales around the mean.
- Variance: Measures the degree of variation in sales.



Biggest Finding

One of the most significant insights from the dataset is the correlation between product categories and sales volume. For instance, the analysis may reveal that certain product categories, such as "Whiskey," consistently generate higher sales volumes and revenue compared to other categories. This could be visualized through a bar chart showing the total sales

Discuss What You Have Learned from the Data and How You Might Use This Knowledge

Key Learnings:

Sales Trends and Customer Preferences:

- The dataset reveals clear sales trends, showing which products are most popular among customers. Understanding these preferences allows for more targeted marketing efforts and better inventory management.

Geographical Insights:

- The data might show that certain store locations or regions contribute more to overall sales, indicating that these areas have a higher demand for specific products. This insight can help in planning regional promotions or stocking strategies.

Impact of Product Pricing:

- By analyzing the relationship between product pricing (e.g., State Bottle Retail) and sales volume, the data can uncover how price changes affect customer purchasing behavior. Products with more elastic demand may require more careful pricing strategies.

Application of Knowledge:

1. **Marketing and Promotions:**
    - Use the insights from product category performance to design targeted marketing campaigns that focus on best-selling items. Seasonal promotions or discounts can also be aligned with the products that have shown high demand during specific times of the year.
2. **Inventory Management:**
    - Implement data-driven inventory management practices, ensuring that high-demand products are always in stock. Avoid overstocking less popular items, which can help reduce storage costs and minimize waste.
3. **Strategic Expansion:**
    - If certain geographic regions show higher sales, consider expanding the product range or opening new stores in those areas. Alternatively, underperforming regions might need different strategies, such as tailored marketing or adjusted pricing.
4. **Pricing Strategy:**

- Utilize the relationship between price and sales volume to refine pricing strategies. For products with elastic demand, consider more dynamic pricing models that respond to market conditions, such as competitor pricing or seasonal demand shifts.